

DRONE



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
12 CLASSES (1 HOUR PER
CLASS)**

**MODE:
OFFLINE**

This is a powered, mechanical and hands-on learning solution focused on drones, where experiments connect textbook concepts to real-world applications. Each session involves building, assembling and testing drone components and complete flying models, both motorised and electronic, to understand how drones work in real-life scenarios. Students learn core STEM concepts such as aerodynamics, propulsion, electronics, sensors and flight control systems through practical activities. High-quality imported drone kits, motors, frames and electronic components made from durable, industry-grade materials are used to ensure safe, reliable and professional learning experiences.

COURSE CURRICULUM

Level-1

Session 1 : Discover the Drone World

- Introduction to drones and their everyday uses (delivery, photography, rescue, agriculture).
- Understanding drones as flying machines.

Session 2 : Anatomy of a Drone

- Frame: Structure that holds all drone components together.
- Motors & Propellers: Components responsible for lift and movement.
- Battery: Power source of the drone.
- Controller & Receiver: Devices used to control the drone wirelessly.

Session 3 : Secrets of Flight

- Basics of lift, thrust, drag and gravity.
- How propellers push air to make the drone fly.

Session 4 : Let's Build – Part 1

- Identifying drone parts and tools.
- Assembling the drone frame and motor placement.

Session 5 : Let's Build – Part 2

- Attaching propellers and battery safely.
- Understanding balance and symmetry in drone building.

Session 6 : Mastering the Controls

- Learning basic drone movements: up, down, left, right, forward and backward.
- Introduction to joystick controls and directions.

Session 7 : Ignition and Inspection

- Pre-flight safety checks.
- Battery check, propeller inspection and safe handling practices.

Session 8 : Into the Skies

- First guided flight experience.
- Understanding take-off, hovering, and landing.

Session 9 : Pilot in Command

- Learning responsibility and rules of safe flying.
- Developing focus, coordination, and confidence while flying.

Session 10 : Aerial Fun Missions – Part 1

- Simple flying challenges like hovering at a spot.
- Path-following and controlled movement activities.

Session 11 : Aerial Fun Missions – Part 2

- Obstacle avoidance and target landing missions.
- Team-based flying activities to improve coordination.

Session 12 : Drone Show and Awards

- Demonstration of learned skills through a mini drone show.
- Recognition and awards for participation and achievements.

Level-2

Session 1: Engineering the Flight

- Understanding advanced drone architecture and system design.
- Exploring professional drone platforms and their core technologies.

Session 2: Building the Core

- Understanding the purpose of major drone components.
- Learning how individual systems work together during flight.

Session 3: Electronic Foundations

- Understanding the electronic systems used in FPV drones.
- Learning power distribution and component integration.

Session 4: Precision Assembly – I

- Understanding the drone frame structure and assembly process.
- Installing motors and essential hardware accurately.

Session 5: Precision Assembly – II

- Completing the installation of electronic components.
- Preparing the drone for wiring and system integration.

Session 6: Propulsion System Testing

- Understanding motor operation and thrust generation.
- Testing motor performance and rotation accuracy.

Session 7: Flight Controller Configuration

- Understanding flight controller software and configuration tools.
- Configuring essential flight parameters and system settings.

Session 8: Flight Controller Calibration

- Calibrating sensors and verifying system accuracy.
- Testing firmware configuration and controller performance.

Session 9: Pre-Flight Validation

- Performing complete system inspection before flight.
- Verifying hardware reliability and operational readiness.

Session 10: Wireless Command System

- Understanding wireless communication between the drone and transmitter.
- Configuring reliable control system connections.

Session 11: Stability Engineering

- Understanding sensor-based flight stabilization.
- Testing gyroscope and accelerometer performance.

Session 12: System Integration

- Integrating hardware, sensors and communication systems.
- Verifying complete drone functionality before operation.

Session 13: Intelligent Flight Control

- Understanding flight stabilization and control algorithms.
- Optimizing drone response for smooth operation.

Session 14: Performance Optimization

- Identifying common performance issues.
- Applying techniques to improve flight efficiency and stability.

Session 15: Advanced Flight Challenges

- Applying flying skills in practical mission-based tasks.
- Improving precision, control and pilot confidence.

Session 16: Capstone Flight Mission

- Demonstrating complete drone building and flying skills.
- Completing a final mission through teamwork and problem-solving.

Level-3

Session 1: The Journey Begins

- Understanding the fundamentals of modern drone technology.
- Exploring the evolution and real-world applications of drones.

Session 2: Inside the Machine

- Understanding the purpose of essential drone systems.
- Exploring how different components work together.

Session 3: Building the Foundation

- Understanding the core electronic systems of a drone.
- Preparing the hardware for successful integration.

Session 4: Shaping the Structure

- Learning professional assembly techniques.
- Constructing the primary framework of the drone.

Session 5: Bringing It Together

- Integrating mechanical and electronic systems.
- Preparing the drone for operational readiness.

Session 6: Power in Motion

- Understanding propulsion and power transmission.
- Verifying system performance through controlled testing.

Session 7: Taking Command

- Understanding wireless communication and control systems.
- Establishing reliable interaction between pilot and drone.

Session 8: Finding Stability

- Understanding how sensors maintain controlled flight.
- Testing system responsiveness and balance.

Session 9: System Sync

- Integrating hardware and software into a complete flight system.
- Verifying coordinated operation across all components.

Session 10: Intelligent Control

- Understanding automated flight stabilization.
- Optimizing system response for smoother performance.

Session 11: Mission Control

- Understanding professional drone configuration tools.
- Configuring essential flight system parameters.

Session 12: Fine-Tuning the System

- Calibrating and optimizing the drone for reliable operation.
- Validating software and hardware integration.

Session 13: Ready for Launch

- Performing complete system verification before operation.
- Ensuring the drone is prepared for practical use.

Session 14: Beyond the Build

- Identifying performance improvements and system refinements.
- Applying solutions to enhance overall reliability.

Session 15: Into the Challenge

- Applying technical knowledge through practical missions.
- Developing precision, confidence and problem-solving skills.

Session 16: The Final Mission

- Demonstrating complete drone engineering and piloting skills.
- Completing a mission through teamwork and strategic execution.

Level-4

Session 1: Flight Systems Revisited

- Revisiting advanced drone architecture and essential flight systems.
- Refreshing assembly knowledge, safety procedures and operational workflow.

Session 2: Flight Readiness

- Performing complete drone inspection, calibration and pre-flight testing.
- Identifying hardware issues and verifying system reliability.

Session 3: Precision Flight

- Refining manual flying skills through guided flight exercises.
- Improving control, orientation and overall flight stability.

Session 4: Performance Analysis

- Diagnosing common flight issues and understanding their causes.
- Applying troubleshooting techniques to improve overall drone performance.

Session 5: Smart Computing Platform

- Understanding onboard computing for autonomous drone applications.
- Setting up the Raspberry Pi development environment.

Session 6: Software Foundations

- Learning Python fundamentals for autonomous drone programming.
- Developing the coding skills required for intelligent flight operations.

Session 7: Environmental Awareness

- Understanding how autonomous drones perceive their surroundings.
- Exploring sensing technologies used for intelligent navigation.

Session 8: Intelligent Inputs

- Integrating sensors with the onboard computing platform.
- Understanding sensor communication and real-time data acquisition.

Session 9: Data Intelligence

- Reading, processing and interpreting sensor information.
- Converting environmental data into actionable decisions.

Session 10: Autonomous Logic

- Understanding decision-making algorithms for autonomous systems.
- Developing intelligent control workflows for drone operation.

Session 11: System Integration

- Combining computing, sensing and flight control into one platform.
- Testing communication between all integrated subsystems.

Session 12: Autonomous Configuration

- Optimizing software and hardware for reliable autonomous operation.
- Validating integrated system performance before live testing.

Session 13: Flight Validation

- Preparing the autonomous system for real-world operation.
- Performing comprehensive pre-flight verification and safety checks.

Session 14: Autonomous Testing

- Evaluating autonomous functions through controlled flight trials.
- Identifying and resolving operational challenges during testing.

Session 15: Intelligent Missions

- Applying autonomous concepts through practical flight scenarios.
- Improving navigation, precision and mission execution.

Session 16: Autonomous Capstone

- Demonstrating complete autonomous drone development skills.
- Showcasing intelligent flight through an end-to-end autonomous mission.

Class Structure

1. Introduction to Drone Components

- Learn about basic drone parts such as motors, propellers, batteries, and controllers.
- Understand their function in making a drone fly.

2. Supervised Drone Assembly

- Build a beginner-friendly drone using high-quality imported kits.
- Assembly done under guidance to ensure safety and correctness.

3. Understanding Flight Concepts

- Simple explanation of aerodynamics and motion.
- Connecting drone flight concepts to real-world flying objects like birds

and airplanes.

4. Practical Flying Activities and Challenges

- Fun challenges including:
- Take-off and landing tasks
- Hover control games
- Direction control missions
- Hands-on flying practice to build confidence

5. Safety, Discipline and Real-Life Applications

- Learning drone safety rules and responsible usage.
- Understanding how drones are used in real-life professions.